is not same as (a,b) as they both indicate different orders.

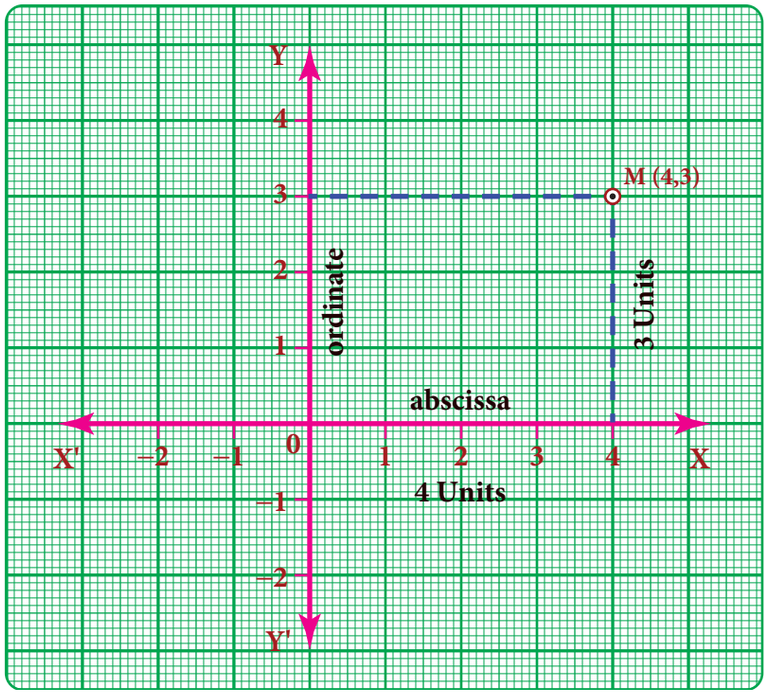
We have, XOX' and YOY' as the co-ordinate axes and let 'M' be (4,3) in the plane.To locate 'M'

(i) you (always) start at O, a *fixed* point (which we have, agreed to call as *origin*),

(ii) *first*, move 4 units along the horizontal direction (that is, the direction of *x*-axis)

(iii) *and then* trek along the *y*- direction by 3 units.

To understand how we have travelled to reach M, we denote by (4,3).



4 is called the *x*-coordinate of M and 3 is called the *y*-coordinate of M.

It is also habitual to name the ***x*-coordinate as abscissa** and the ***y*-coordinate as ordinate**. (4,3) is as an *ordered pair*.

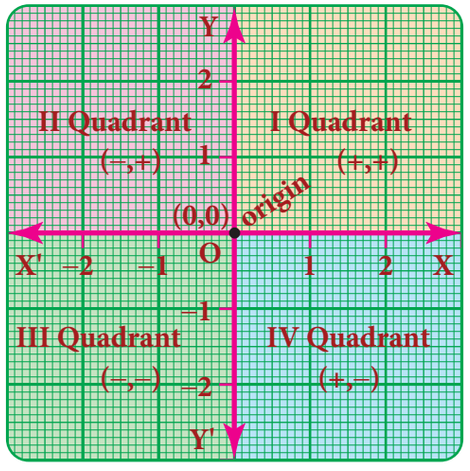
Think

If instead of (4,3), we write (3,4) and try to mark it, will it represent 'M' again?

3.9.5 Quadrants

The coordinate axes divide the plane of the graph into four regions called quadrants. It is a convention that the quadrants are named in the anti clock wise sense starting from the positive side of the X axis.

Quadrant	Sign
I the region XOY	$x>0$, $y>0$, then the coordinates are (+,+) Examples: (5,7) (2,9) (10,15)
II the region X'OY	$x<0$, $y>0$ then the coordinates are (−,+) Examples (−2,8) (−1,10) (−5,3)
III the region X'OY'	$x<0$, $y<0$ then the coordinates are (−,−) Examples: (−2,−3) (−7,−1) (−5,−7)
IV the region XOY'	$x>0$, $y<0$ then the coordinates are (+,−) Examples : (1,−7) (4,−2) (9,−3)



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